Your grade: 87.50%

Your latest: **87.50%** • Your highest: **87.50%** • To pass you need at least 80%. We keep your highest score.

Next item

1. When designing the tools for an AI agent, why is it important to provide precise and clear descriptions for each function? Select the best answer. — 1 / 1 point

- Because precise descriptions automatically reduce the number of tokens used in the conversation loop.
- Because clear descriptions allow the model to bypass the orchestration layer and talk directly to the user.
- Because the descriptions are displayed directly to the end user as a manual for the application.
- Because the model relies on descriptions to understand exactly when and how to use the tool for a specific task.

Nice work

Correct! If a tool is described incorrectly (like describing a hammer as a screwdriver), the model will attempt to use it the wrong way, leading to a failure in the agent's logic.

2. A developer is inconsistently receiving high-quality summaries from an AI model. Following a structured evaluation workflow, what should be their first step to transform this unpredictable behavior into a repeatable skill? Select the best answer. — 1 / 1 point

- Define specific success criteria, such as accuracy, tone, and word count limits, to measure output quality.
- Add the current inconsistent prompt to a centralized prompt library for the rest of the team to use.
- Run an A/B test between Gemini Pro and Gemini Flash to see which model is faster.
- Create a complex prompt template including Role, Context, Constraints, and Output.

Nice work

Correct! The first step in any evaluation is to define success. Without clear, measurable standards, it is impossible to determine if a prompt or model is actually performing well.

3. When a developer goes from a simple baseline prompt to a structured template, which of the following components is specifically responsible for defining the boundaries the AI must work within, such as a 50-word limit? Select the best answer. — 0 / 1 point

- Context
- Output Format
- Role
- Constraints

Try again

Not quite. Please refer to *From static to dynamic AI.*

4. When you execute a function-calling script, what typically happens to the response text when the AI model decides that a tool is needed to answer a user's request? Select the best answer. — 1 / 1 point

- The model will raise a value error if it detects that the user's natural language request contains a currency ID.
- The response text provides a step-by-step mathematical explanation of the exchange before the tool is called.
- The response text is automatically saved as a PDF file in the tools_exchange.py directory.
- The model returns a structured object containing the function name and an argument dictionary in its place.

Nice work

Correct! When Gemini determines a tool is needed, it pauses the text generation and sends a "function call" request. The developer then has to execute that function and send the results back to get the final text response.

5. In a multi-capability agent, what is the advantage of using Google Search grounding when you ask a model about a current technical topic? Select the best answer. — 0.8 / 1 point

- [x] It provides a way to get accurate, real-time answers from the live web without the developer having to manually integrate a separate search API.

Nice work

Correct! Grounding acts as a built-in bridge to the live web. It allows the model to find fresh information and cite sources automatically, making the final response more reliable and timely.

- [ ] It converts the model's natural language response into a high-speed video generation format for terminal viewing.
- [ ] It allows the model to summarize a specific, known article by reading only the URL provided by the developer.
- [ ] It forces the model to ignore its own internal training data and only provide a list of recommended actions.

You didn't select all the correct answers

6. Once an AI product is live and accessible to the public, what is the purpose of implementing usage limits and continuous monitoring systems? Select the best answer. — 1 / 1 point

- They allow the application to automatically rewrite its own production plan every time a new user signs up.
- They ensure the application only functions during standard business hours to save on electricity.
- They prevent the AI from generating any response that exceeds five words in length.
- They protect the business from unexpected spikes in activity and maintain visibility into real-time performance and costs.

Nice work

Correct! Monitoring and usage controls provide a safeguard against sudden surges in traffic that could lead to reliability issues or unexpectedly high expenses, ensuring the product remains cost-effective over time.

7. When using an AI model to analyze an uploaded image file, such as a JPEG, how is the cost, or token consumption, of that action determined? Select the best answer. — 1 / 1 point

- The image analysis is always free of charge as long as the user does not ask the AI to summarize the results in a poem.
- The cost is based solely on the number of words the AI writes when describing what it sees in the image.
- The system applies a base token amount for the image, which can increase depending on factors like the resolution or depth of the analysis.
- The token count for images is calculated by counting the number of physical pixels and multiplying them by the current cost of a "Pro" model.

Nice work

Correct! Image features typically involve a base token overhead. When the model performs the analysis, the total token count increases by an amount related to the image's complexity and the requested level of detail.

8. A startup is using Google Cloud Run to deploy their AI application. What is a benefit of using this serverless approach when the company is dealing with unpredictable user traffic? Select the best answer. — 1 / 1 point

- It automatically adjusts the application, so it behaves differently on a local machine than it does in the cloud to make sure any testing stays private.
- It requires the startup to only purchase and install additional server racks when they reach one thousand users, two thousand users, and so on.
- It automatically scales resources up to handle high traffic and down to zero during quiet periods so the user only pays for what is used.
- It means the startup won't need to use an API key because it will use the cloud's built-in security firewall.

Nice work

Correct! This automatic scaling makes deployment highly cost-efficient and reliable, as it adjusts to user demand without requiring manual configuration or intervention.

9. A developer you are working with asks, "What is the role of billing alerts in the long-term maintenance of a scalable AI application?" Which of the following answers would you give her? Select the best answer. — 1 / 1 point

- They provide a daily report on the tokens each user is generating so they can be billed appropriately.
- They serve as an early warning system that notifies developers when spending approaches a set limit.
- They increase the cost per token for the application whenever the apps reserve account gets too low.
- They automatically block users from a specific country if the exchange rate for tokens becomes too high.

Nice work

Correct! Billing alerts help manage the financial health of a project as it scales, preventing unexpected expenses and ensuring the application stays within its intended budget.

10. A developer asks, "Why do we establish threshold rules and triggers in a cloud budget alert system?" Which of the following answers would you give them in reply? Select the best answer. — 1 / 1 point

- To send notifications, such as emails, when the application's actual or forecasted spending reaches a percentage of a target amount.
- To increase the Target Amount of the budget every time the application's summary reduction percentage drops below 50%.
- To automatically translate the application into different languages once a certain number of tokens are used.
- To force the AI model to switch from High thinking level to Low thinking level to control costs whenever the monthly budget is met or exceeded.

Nice work

Correct! Threshold rules act as a warning system. By setting a target amount like $5 US, you can be notified if usage spikes, helping you manage costs and avoid unexpected bills.